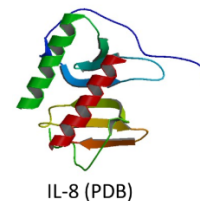


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🌐 Sandwich ELISA for IL-8 detection in supernatants

DOI

<https://dx.doi.org/10.17504/protocols.io.gz7bx9n>



Stefan Odenbreit¹, Luisa F. Jiménez-Soto¹

¹Max von Pettenkofer-Institut für Hygiene und Medizinische Mikrobiologie, Ludwig-Maximilians-Universität



Luisa F Jimenez-Soto

Walther Straub Institute for Pharmacology and Toxicology

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Abstract

The IL-8 ELISA protocol here described was established by Stephan Odenbreit and modified in this form by Luisa F. Jiménez-Soto.

Here are the publications using this protocol:

DOI:[10.1371/journal.pone.0035341](https://doi.org/10.1371/journal.pone.0035341)

DOI:[10.1128/IAI.00364-09](https://doi.org/10.1128/IAI.00364-09)

DOI:[10.1078/1438-4221-00205](https://doi.org/10.1078/1438-4221-00205)

and PMID:11886563

Materials

STEP MATERIALS

- ⊗ Purified Mouse Anti-Human IL-8 **Becton Dickinson (BD) Catalog #554716**
- ⊗ Fetal Bovine Serum **Gibco - Thermo Fisher Scientific Catalog #10270106**
- ⊗ VECTASTAIN ABC HRP Kit (Peroxidase, Standard) **Vector Laboratories Catalog # PK-4000**
- ⊗ TMB Substrate Reagent Set **Becton Dickinson (BD) Catalog #555214**
- ⊗ Biotin Mouse Anti-Human IL-8 **Becton Dickinson (BD) Catalog #554718**
- ⊗ Purified Mouse Anti-Human IL-8 **Becton Dickinson (BD) Catalog #554716**
- ⊗ Fetal Bovine Serum **Gibco - Thermo Fisher Scientific Catalog #10270106**
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Coating of plates

- 1
 - Resuspend 30µl (15 µg) coating antibody (Anti-IL-8 Antibody IL-8 Human 554716 Pharmingen, 0,5µg/µl) in 5 ml Coating buffer (100 mM Na₂HPO₄ pH 9,6)
 - Pipette 50µl of antibody solution in each well of a Nunc MaxiSorp® flat-bottom 96 well plate. Take care to cover the whole surface of each well with the solution. Incubate at 4°C for minimum 8 hours

 Purified Mouse Anti-Human IL-8 **Becton Dickinson (BD)** Catalog #554716

Remove unbound antibody

- 2 Remove the unbound antibody, washing twice with 200 µl/well washing buffer (PBS 0,05% Tween 20 (v/v))

Blocking the plate

- 3 Add 100µl to each well a PBS 10% FBS solution for blocking, for minimum 1 hour at 37°C or 2 hours at room temperature.

 Fetal Bovine Serum **Gibco - Thermo Fisher Scientific** Catalog #10270106

Remove blocking solution

- 4 Remove blocking solution and wash well twice with 200µl Wash buffer per well.

Prepare standards and samples and add to plate

- 5
 - Add standard solutions (100µl / well)
 - Prepare samples by mixing thoroughly before loading them onto the plate. (final volume per well: 100 µl).

Incubate samples

- 6 Incubate samples for minimum 5 hours at 37°C, or overnight at 4°C. Do not shake.

Wash to remove unbound molecules

- 7 Remove thoroughly the samples by washing 6 times with 200 µl wash buffer / well.

Add the detection antibody

- 8 Add 100µl/well of a solution of 10µg biotinylated IL-8 antibody (BD Biosciences, Catalog No.554718) in 10 ml wash buffer 10% FBS, and incubate for minimum 45 min at 37°C, or 2 hours at room temperature.

 Biotin Mouse Anti-Human IL-8 **Becton Dickinson (BD) Catalog #554718**

Prepare the Peroxidase (POX)-Avidin complex

- 9 30 minutes before the incubation of the detection antibody is ready, prepare the POX - Streptavidin complex, by mixing 1,5µl of solution A and 1,5µl of solution B (Vectastain ABC kit, Catalog Number: PK-4000) in 200µl ELISA buffer . Vortex shortly to mix and incubate for 30 min at room temperature.

 VECTASTAIN ABC HRP Kit (Peroxidase, Standard) **Vector Laboratories Catalog # PK-4000**

Remove unbound biotinylated antibody

- 10 Once the incubation of the biotinylated antibody is ready, remove all unbound antibody from plate by washing 6 times as described before

Add the POX-Streptavidin solution (detection complex)

- 11 Prepare the POX-Streptavidin solution by diluting the 200 µl previously prepared in 10 ml of ELISA buffer (50 mM TRIS-HCL pH 7,6). MIX BY INVERSION ONLY. Do not vortex.
- 12 Add 100µl of the POX-Strep complex solution in each well. Incubate for 45 min at room temperature. Do not vortex the POX-Step complex.
- 13 Remove the unbound complex, by washing as previously described 6 times. Develop using a TMB substrate (5 ml Solution A + 5 ml Solution B), 100 µl per well, and stop reaction using 50 µl of a 1M H₂SO₄. Read in Wavelength described by TMB substrate instructions.

 TMB Substrate Reagent Set **Becton Dickinson (BD) Catalog #555214**